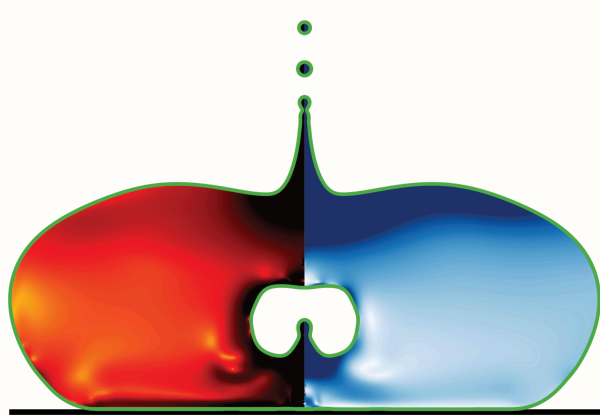


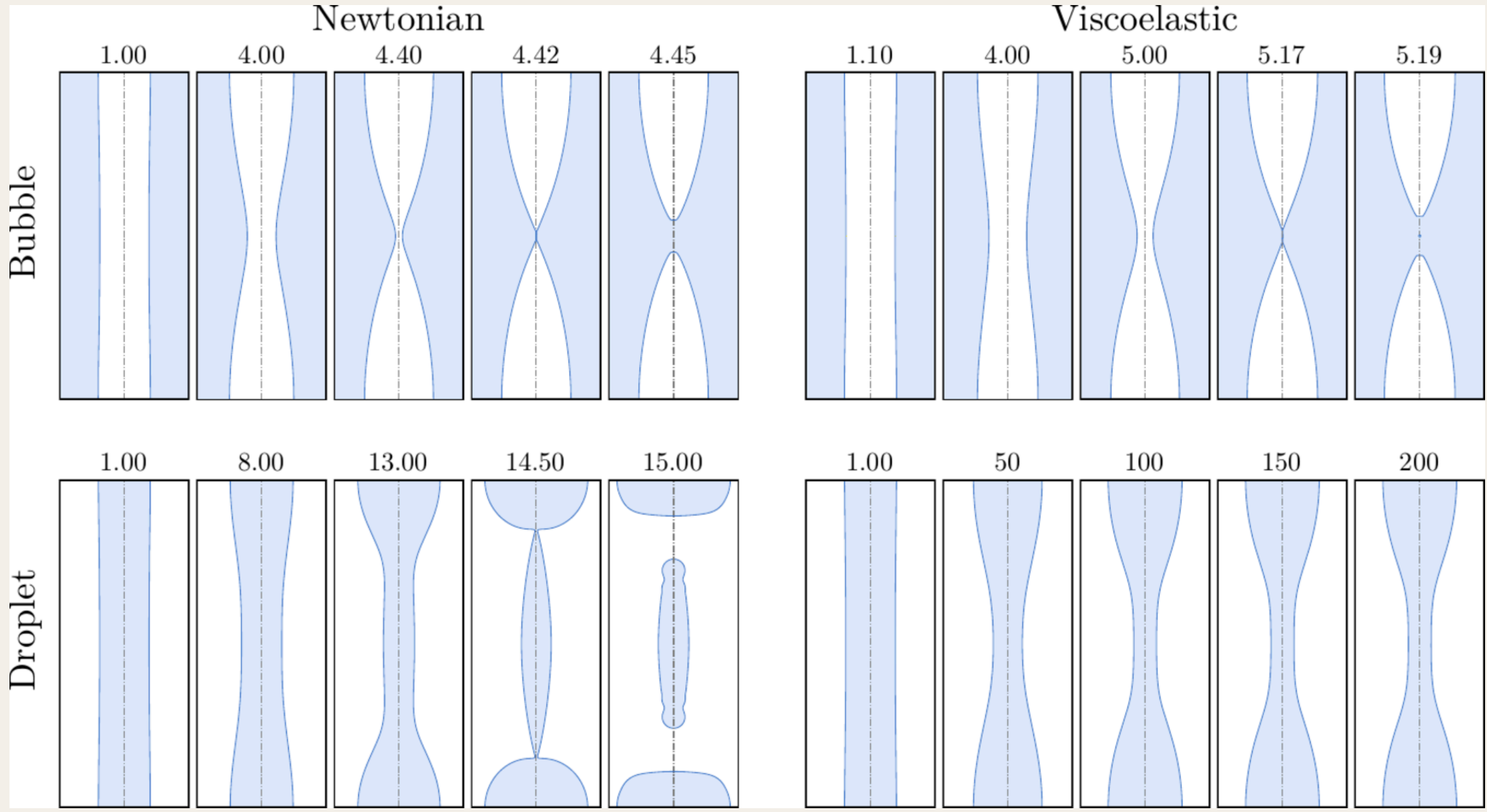
Hydrodynamic singularities

When a free surface pinches off, the flow forgets its past. A pinch of polymer rewrites the ending — but only for a drop, not a dilute bubble.



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Direct numerical simulation (**Basilisk C**, Oldroyd-B). Top — a **bubble** neck pinches off; Newtonian (left) and viscoelastic (right) look alike. Bottom — a **drop** neck; Newtonian (left) pinches, but the same polymer (right) draws out a long, persistent thread.

What is a singularity?

free-surface flow

pinch-off

self-similarity

DNS · Basilisk

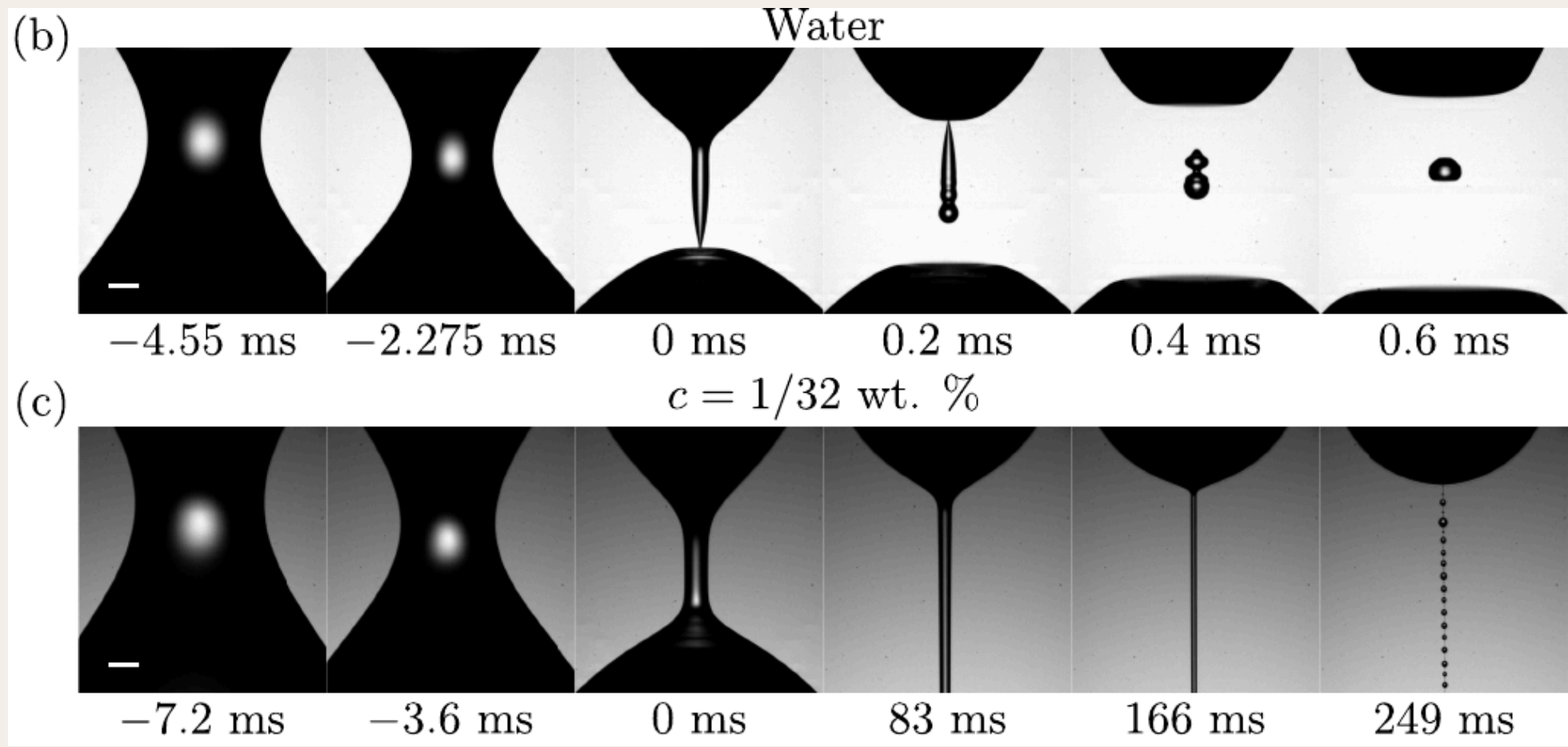
Pinch-off, coalescence, the snap of a sheet or jet — soft-matter flows routinely concentrate geometry, stress and time into a single point, where the neck radius, the curvature and the strain rate diverge together.

Near pinch-off the flow loses memory of how it began: the neck thins self-similarly, $h \sim (t_0 - t)^\alpha$, with $\alpha = 2/3$ for a Newtonian drop and $\alpha \simeq 1/2$ for a bubble.

Drops: a thread arrests pinch-off

A Newtonian drop reaches a finite-time singularity. Add a dilute polymer and the chains stretch along the neck, building an elastic stress that diverges strongly, $\sigma_{zz} \sim G(h_0/h)^4$, and halts the collapse.

The neck survives as a slender filament — the classic **beads-on-a-string**.



Drop pinch-off. Top: a Newtonian neck thins to a true finite-time singularity. Bottom: a dilute polymer turns the neck into a persistent beads-on-a-string filament.

→ Dilute polymer → a thread; pinch-off arrested

Why elasticity picks sides

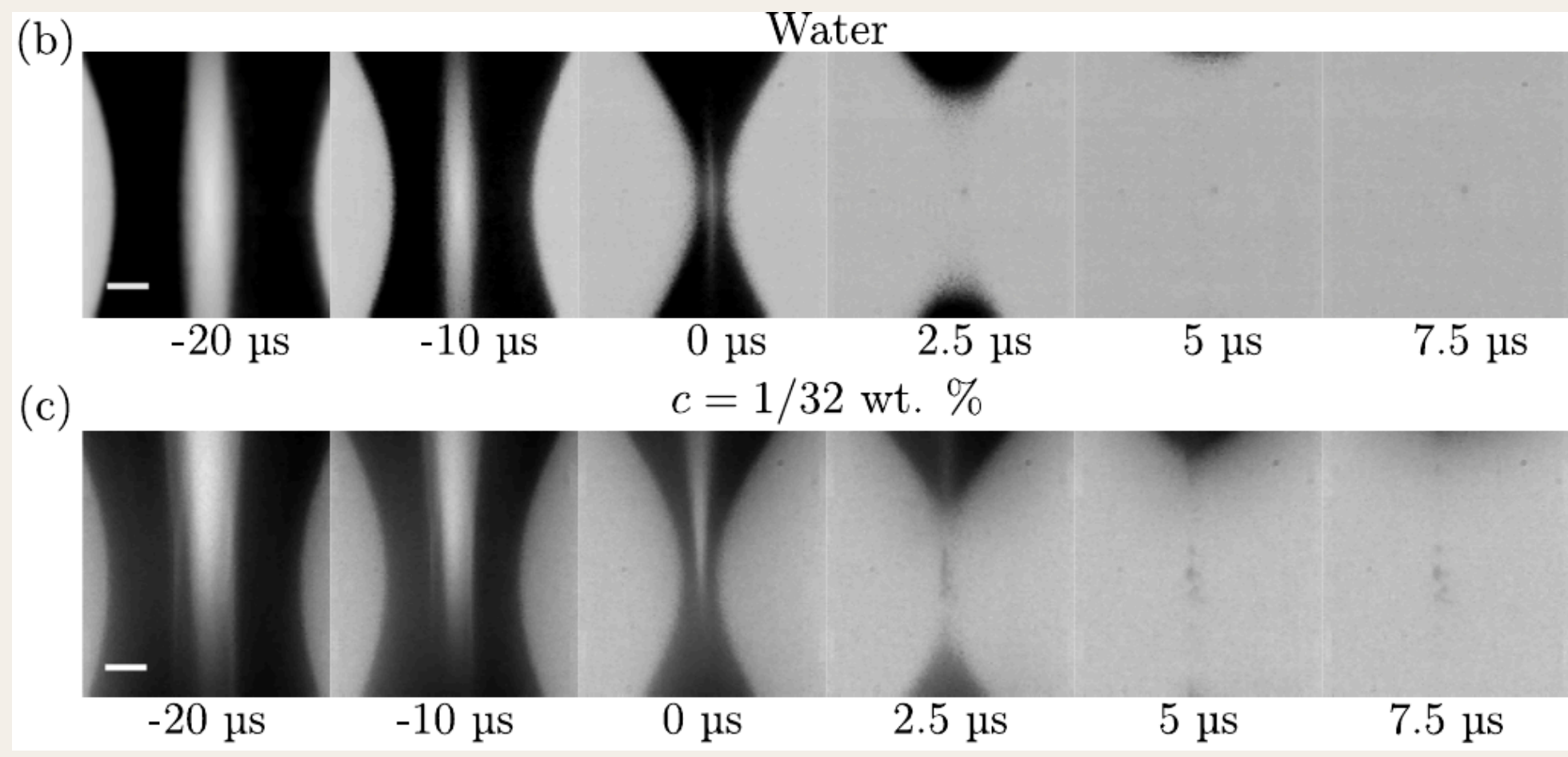
$$\underbrace{\sigma_{zz} \sim G\left(\frac{h_0}{h}\right)^4}_{\text{drop}} \gg \underbrace{\sigma_{rr} \sim G\left(\frac{h_0}{h}\right)^2}_{\text{bubble}}$$

Both stresses are singular at pinch-off, but the bubble's diverges far more weakly — a drop's balance is axial and capillary, a bubble's is radial and inertial. So a dilute polymer grows a thread on a drop, while a bubble pinches like a Newtonian fluid until the elastocapillary number $\text{Ec} = Gh_0/\gamma$ is pushed up by high concentration.

Bubbles: the singularity survives

A bubble neck pinches just as sharply, $h \sim (t_0 - t)^{1/2}$, but the weak stress cannot organise a thread in a dilute solution — the pinch-off stays Newtonian-like.

A thread appears only at **high polymer concentration**, where it becomes sensitive to the size of the needle the bubble detaches from.



Bubble pinch-off. Top: Newtonian. Bottom: viscoelastic. In the dilute limit the two near-singular sequences are almost indistinguishable.

→ Dilute polymer → no thread; pinch-off survives

What it tells us

- One polymer additive, two free surfaces, **opposite fates**: a thread arrests the drop, while the dilute bubble pinches like a Newtonian fluid.
- The split is set by how strongly the elastic stress diverges near the singularity — as $(h_0/h)^4$ for a drop versus $(h_0/h)^2$ for a bubble.
- Captured by direct simulation and an Oldroyd-B model. Singularities are where flows turn universal; knowing which ones elasticity can tame guides how we control breakup, sprays and aerosols.

References

[1] Verschuur, C. I., Oratis, A. T., Sanjay, V. & Snoeijer, J. H. How elasticity affects bubble pinch-off. *arXiv:2511.20075* (2026); submitted to *Phys. Rev. Fluids*.
[2] Eggers, J. Nonlinear dynamics and breakup of free-surface flows. *Rev. Mod. Phys.* **69**, 865 (1997).
[3] Day, R. F., Hinch, E. J. & Lister, J. R. Self-similar capillary pinchoff of an inviscid fluid. *Phys. Rev. Lett.* **80**, 704 (1998).
[4] Burton, J. C., Waldrep, R. & Taborek, P. Scaling and instabilities in bubble pinch-off. *Phys. Rev. Lett.* **94**, 184502 (2005).

[5] Clasen, C. et al. The beads-on-a-string structure of viscoelastic threads. *J. Fluid Mech.* **556**, 283 (2006).
[6] Eggers, J., Herrada, M. A. & Snoeijer, J. H. Self-similar breakup of polymeric threads as described by the Oldroyd-B model. *J. Fluid Mech.* **887**, A19 (2020).
[7] Popinet, S. An accurate adaptive solver for surface-tension-driven interfacial flows. *J. Comput. Phys.* **228**, 5838 (2009). basilisk.fr.

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